Table 11-1. *H-Bridge Function*

Input 1	Input 2	Function
HIGH	LOW	Spin motor clockwise
LOW	HIGH	Spin motor counter-clockwise
LOW	LOW	Disable motor / free spin
HIGH	HIGH	Motor brake / full stop

By writing a `HIGH` signal on pin 8 and a `LOW` signal on pin 9, then the motor will spin one direction and will reverse direction when pin 8 is `LOW` and pin 9 is `HIGH`. If both pins are `LOW`, then the motor will have 0 voltage and will spin freely—assuming it's not a gear motor. If both pins are set to `HIGH`, then current will be applied to both sides of the motor, locking the motors position just like an electronic brake.

From here, all you need to do is set the direction using pins 8 and 9, then control the speed with `analogWrite()` on Pin 10, and you're up and running with an H-bridge.

Relays

In this project, we will briefly look at using a mechanically switching relay to handle high currents and switch directions with a DC motor. Relays work a little like the light switches in our walls, with little metal contacts inside the switch, which are physically moved whenever the switch is activated, completing a circuit. This is done by sending a digital signal to a little magnetic coil inside the relay that activates the switch for us. In Figures 11-8 and 11-9 we are using the Axicom D2N V23105 relay with a +5v coil that can control a motor of up to 3 amps at a ridiculously high voltage of +220 volts.

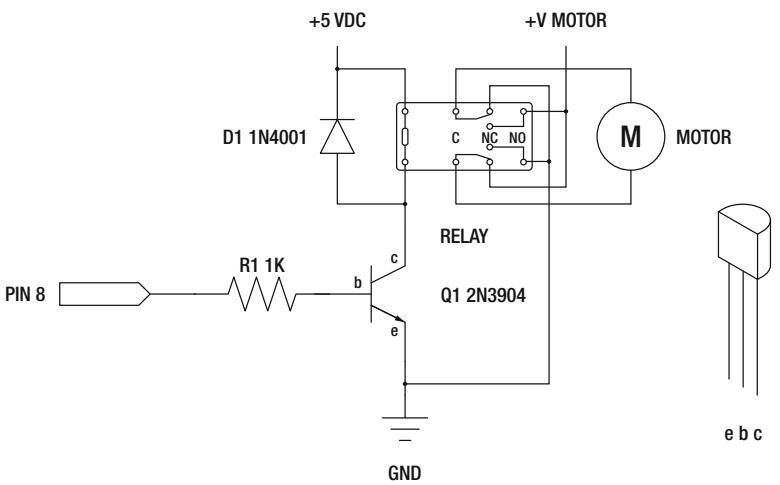


Figure 11-8. *Relay and DC Motor schematic*